Expectation Decider

Insights Report — Theory Edition

Derivable Judgement: A Statistical Decision-Making Model

Dataset: 450 Students | 7 Probability Topics | Theory Part A

Objective: To apply inferential statistics to evaluate multiple hypotheses from a public health dataset — testing theoretical understanding and practical application of hypothesis testing, confidence intervals, statistical errors, z/t/chi-square/ANOVA tests, correlation, and covariance.

1. Inferential Statistics

Inferential statistics is the branch of statistics that draws conclusions about a large population by analyzing a smaller, representative sample. Instead of studying every individual, we use mathematical techniques — probability, estimation, and hypothesis testing — to make predictions and decisions about the whole group with a quantified level of confidence.

Term	Definition
Population	The entire group being studied (e.g., all patients in a region).
Sample	A subset drawn from the population for analysis.
Inference	Using sample results to make probabilistic statements about the population.
Key Tools	Hypothesis tests, confidence intervals, p-values, critical values.

2. Hypothesis Testing & Its Components

Hypothesis testing is a formal statistical procedure used to decide whether there is sufficient evidence in a sample to support a specific claim about the population. It prevents us from making decisions based purely on guesswork — instead, the data drives the conclusion.

Key Components

- H_0 (Null Hypothesis) — The default assumption: 'No effect / No difference.' Example: Smoking has NO effect on Diabetes.
- H_1 (Alternative Hypothesis) — The claim to prove: 'There IS an effect.' Example: Smoking DOES affect Diabetes.
- α (Significance Level) — Acceptable risk of being wrong. Usually set at 0.05 (5%).
- Test Statistic — A calculated number (z, t, χ^2, F) derived from the sample data.
- p-value — Probability that the observed result occurred by chance if H_0 is true.
- Decision Rule — If p-value $< \alpha \rightarrow$ Reject H_0; If p-value $\geq \alpha \rightarrow$ Fail to Reject H_0.

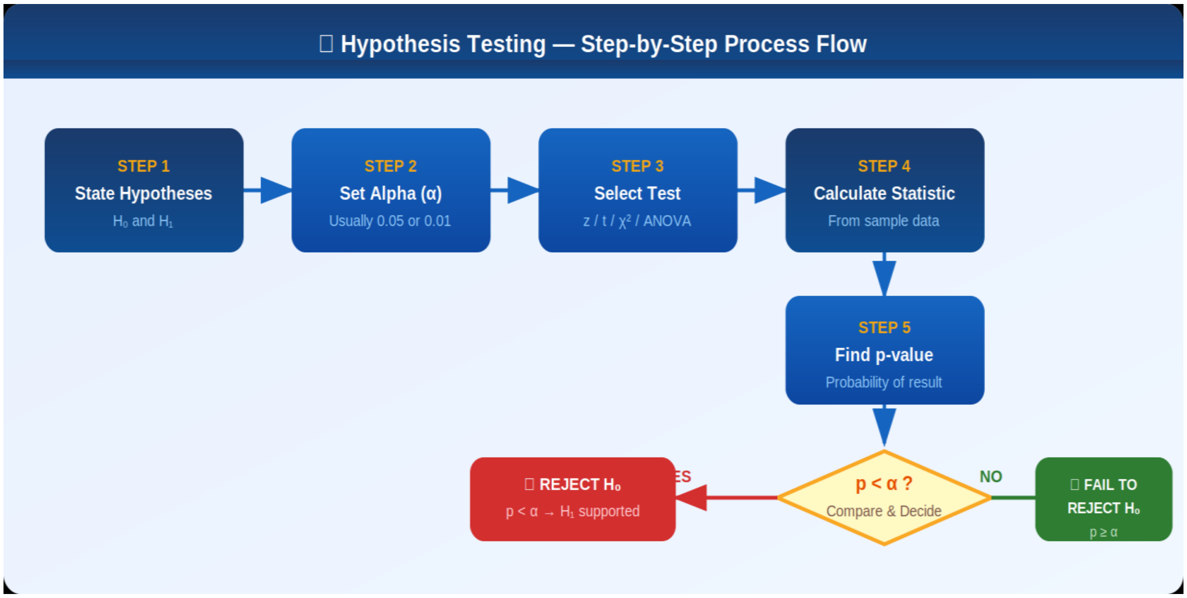

Figure 1 — Hypothesis Testing: Step-by-Step Process Flow

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3. Confidence Interval & Critical Value

Confidence Interval (CI)

A Confidence Interval is a range of values, calculated from sample data, that is likely to contain the true population parameter with a stated probability (e.g., 95%). Rather than giving a single point estimate, CI gives a range that accounts for sampling variability.

Formula (for large samples):

$$CI = \bar{x} \pm Z\,\alpha/2 \times (\sigma / \sqrt{n})$$

Where: $\bar{x}$ = Sample Mean | σ = Standard Deviation | n = Sample Size | $Z\,\alpha/2$ = Critical Value

Interpretation: A 95% CI means: if we repeated this study 100 times, approximately 95 of the resulting intervals would contain the true population mean.

Critical Value

The critical value is the cut-off point on the distribution that separates the 'reject H_0' region from the 'fail to reject H_0' region. It is determined by the significance level (α) and test type. For a two-tailed test at $\alpha = 0.05$, the critical Z-value is ±1.96.

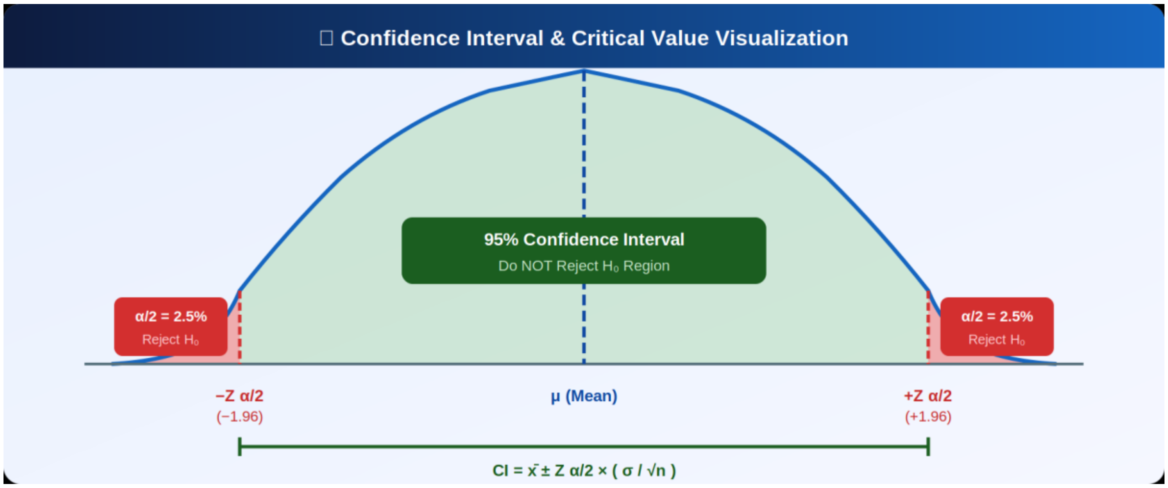

Figure 2 — Confidence Interval & Critical Value on Normal Distribution Curve

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4. P-Value

The p-value (probability value) is the probability of observing a test result as extreme as the one obtained, assuming the null hypothesis (H_0) is true. A small p-value indicates that the observed result is unlikely to have occurred by random chance alone.

Term	Definition
p-value < 0.05	Strong evidence against H_0 → Reject H_0 (statistically significant)
p-value ≥ 0.05	Insufficient evidence → Fail to Reject H_0 (not significant)
p-value < 0.01	Very strong evidence → Reject H_0
p-value = 0.03	Example: 3% chance of seeing this result if H_0 were true

5. Type I and Type II Errors

Even when hypothesis testing procedures are followed correctly, two types of errors can occur. Recognizing these helps in setting appropriate significance levels and correctly interpreting test results.

	Type I Error (α — False Positive)	Type II Error (β — False Negative)
Definition	Rejecting a true H_0	Failing to reject a false H_0
Also Called	False Positive	False Negative
Risk	Claiming effect when none exists	Missing a real effect
Control Via	Lower significance level (α)	Increase sample size / power
Example	Diagnosing disease in healthy person	Missing actual disease in patient

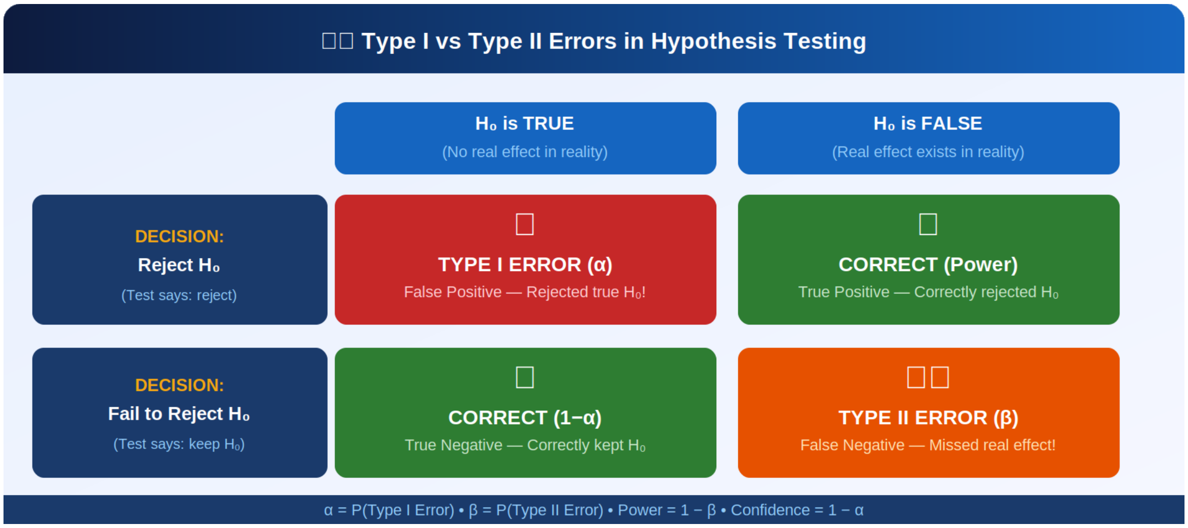

Figure 3 — Type I vs Type II Error: Decision Matrix

6. Statistical Tests: Z, T, Chi-Square & ANOVA

Z-Test

Used when the sample size is large ($n \geq 30$) and the population standard deviation (σ) is known. Compares a sample mean to a population mean to detect significant differences.

Z-Test Formula:

$$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

T-Test

Used when the sample is small ($n < 30$) or population σ is unknown. It uses the t-distribution (heavier tails than normal), which is more appropriate for small samples. Can also compare means of two independent groups.

T-Test Formula:

$$t = (\bar{x} - \mu) / (s / \sqrt{n}) \qquad df = n - 1$$

Chi-Square Test (χ^2)

Used to examine the association between two categorical variables. It compares observed frequencies in a contingency table to expected frequencies under the assumption of no association (H_0).

Chi-Square Formula:

$$\chi^2 = \Sigma [(O - E)^2 / E] \qquad O = \text{Observed}, \quad E = \text{Expected}$$

ANOVA (Analysis of Variance)

Used to compare the means of three or more groups simultaneously. Instead of running multiple t-tests (which inflates error risk), ANOVA uses one F-test to determine if at least one group mean is significantly different.

ANOVA F-Statistic:

$$F = \text{Variance Between Groups} / \text{Variance Within Groups}$$

Decision: If F-statistic > Critical F value (or $p < \alpha$) → At least one group mean is significantly different → Reject H_0.

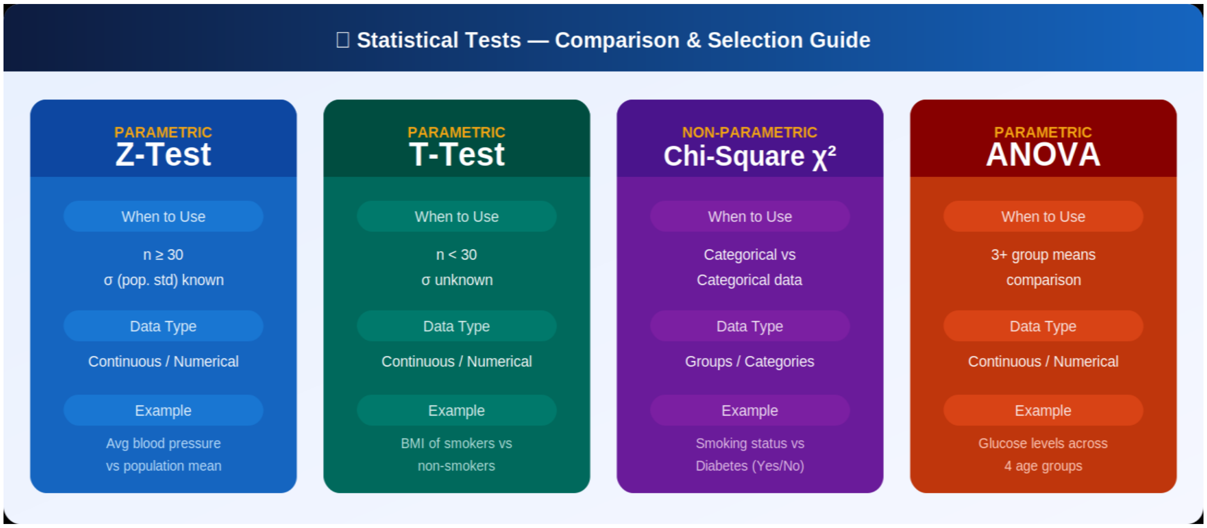

Figure 4 — Statistical Tests: Colour-Coded Comparison Guide

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7. Covariance

Covariance measures how two variables change together. It indicates the direction of the linear relationship — whether they tend to increase together (positive) or move in opposite directions (negative). However, its magnitude depends on the units used, making it hard to compare across datasets.

Covariance Formula:

$$\text{Cov}(X, Y) = \Sigma \left[(X_i - \bar{X})(Y_i - \bar{Y}) \right] / (n - 1)$$

Interpreting Covariance

- Positive Covariance (+) → Both variables move in same direction → e.g., Age & BMI
- Negative Covariance (−) → Variables move in opposite directions → e.g., Exercise & BMI
- Zero Covariance (≈ 0) → No linear relationship between the variables

Limitation: Covariance is scale-dependent — it changes if units change (kg vs lbs). That's why we normalize it into Correlation (r), which is always between −1 and +1.

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8. Correlation

Correlation is a standardized, dimensionless measure of the strength AND direction of the linear relationship between two continuous variables. The Pearson Correlation Coefficient (r) is the most common and always ranges from −1 to +1.

Pearson Correlation Formula:

$$r = \text{Cov}(X, Y) \,/\, (S_x \times S_Y) \qquad \text{Range: } -1 \le r \le +1$$

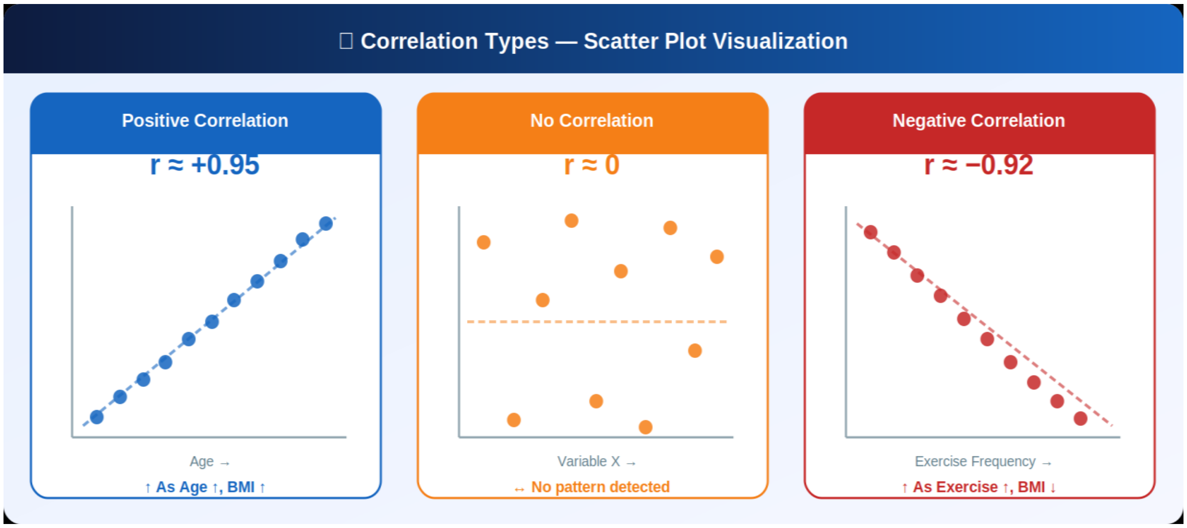

Figure 5 — Correlation Types: Positive, None, and Negative Scatter Plots

Interpreting r Values

- $r = +1.0$ → Perfect positive correlation (both variables rise together exactly)
- $r = 0.7$ to 0.9 → Strong positive correlation
- $r = 0.4$ to 0.6 → Moderate positive correlation
- $r \approx 0.0$ → No linear correlation (random scatter)
- $r = -0.4$ to -0.9 → Moderate to strong negative correlation
- $r = -1.0$ → Perfect negative correlation

Covariance vs Correlation: Covariance tells direction only; Correlation tells both direction AND strength, on a standardized scale — making it easy to compare relationships across any variables in the dataset.

★ Quick Reference Summary

#	Concept	Core Idea	Formula / Key Metric	Project Use
1	Inferential Stats	Draw conclusions about population from sample	Probability + Estimation	All analyses
2	Hypothesis Testing	Test a claim using sample data	H_0, H_1, α, p-value	Smoking vs Diabetes
3	Confidence Interval	Range containing true parameter	$\bar{x} \pm Z \cdot (\sigma/\sqrt{n})$	Age, Weight CI
4	P-value	Probability of result under H_0	$p < \alpha$ → Reject H_0	All tests
5	Type I / II Errors	Wrong rejection / Wrong acceptance	$\alpha = P(\text{Type I})$	Interpretation
6	Z / T Test	Compare means	$Z=(\bar{x}-\mu)/(\sigma/\sqrt{n})$	Blood pressure, BMI
6	Chi-Square	Test categorical	$\Sigma(O-E)^2/E$	Smoking vs Disease

#	Concept	Core Idea	Formula / Key Metric	Project Use
6	ANOVA	Compare 3+ group means	F = VarBetween/VarWithin	Age groups vs Glucose
7	Covariance	Direction of relationship	$\Sigma(X-\bar{X})(Y-\bar{Y})/(n-1)$	Age vs BMI
8	Correlation	Strength + direction (standardized)	$r = Cov/(Sx \cdot Sy)$	Age vs BMI (r value)